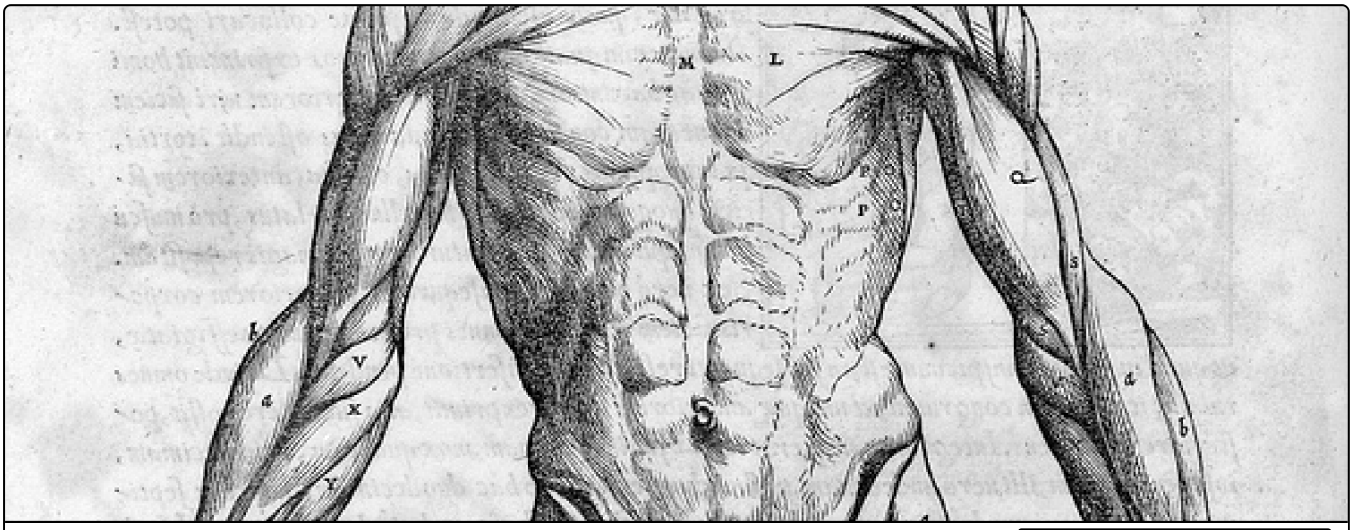
NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 2 CHRONOLOGICAL ENQUIRY SEQUENCE • C1500–C1700

The Medical Renaissance in Britain, c1500–c1700

c1500–c1700: Humanism, Anatomy, Circulation, Scientific Observation & The Great Plague



HISTORICAL PRIMARY RECORD • 1543 CONTEMPORARY PRINT ARCHIVE

Title Page of Andreas Vesalius, De Humani Corporis Fabrica (1543)

Woodcut title page showing Andreas Vesalius conducting an open human dissection surrounded by students, challenging Galenic anatomy directly from empirical observation.

PRIMARY VISUAL EVIDENCE WELLCOME COLLECTION HISTORICAL ARCHIVE

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 2 CONTENT OFFICIAL SPECIFICATION DIRECTORY

1 Ideas About the Cause of Disease and Illness

- Continuity and change in explanations of the cause of disease and illness. A scientific approach, including the work of Thomas Sydenham in improving diagnosis. The influence of the printing press and the work of the Royal Society on the transmission of ideas.

2 Approaches to Prevention and Treatment

- Continuity and change in approaches to prevention, treatment and care in the community and in hospitals.
- Improvements in medical training and the influence in England of the work of Vesalius.

3 Case Studies

- Key individual: William Harvey and the discovery of the circulation of the blood.
- Dealing with the Great Plague in London (1665): approaches to treatment and attempts to prevent its spread.

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

c. 1440–1476 • The Printing Press: Gutenberg & William Caxton

Key Topic 2.1

Johannes Gutenberg invents movable metal type in Mainz; William Caxton establishes England's first press at Westminster in 1476. Medical texts can now be mass-produced with identical diagrams, shattering the Catholic Church's clerical monopoly on manuscript copying.

1543 • Andreas Vesalius Publishes De Humani Corporis Fabrica

Key Topic 2.3

At the University of Padua, professor Andreas Vesalius carries out human dissections personally. The Fabric of the Human Body features precise woodcuts disproving over 200 of Galen's errors, proving the human jawbone is one piece and the breastbone has three segments.

1628 • William Harvey Publishes De Motu Cordis

Key Topic 2.4

Royal physician William Harvey publishes On the Motion of the Heart. Through dissection of cold-blooded reptiles and mathematical calculations, he proves that the heart acts as a mechanical pump recirculating blood continuously through one-way pocket valves.

1660 • Foundation of the Royal Society & Nullius in Verba

Key Topic 2.1

The Royal Society is founded at Gresham College, London, receiving a royal charter from Charles II in 1662. Adopting the motto Nullius in Verba ('Take nobody's word for it'), it launches Philosophical Transactions in 1665 to publish and peer-review scientific experiments.

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two.

Living Timeline Dual-Coding Protocol: Draw visual symbols for clinical bedside observation, London pest houses, microscope animalcules, and Sydenham's disease classification across the timeline below.

1661 • Marcello Malpighi Observes Capillaries Under the Microscope

Key Topic 2.4

Using an early compound microscope, Italian biologist Marcello Malpighi observes blood passing through microscopic capillaries connecting arteries to veins in a frog's lungs. This provides the definitive optical proof for William Harvey's circulatory model.

1665 • The Great Plague of London: Red Crosses & Pest Houses

Key Topic 2.5

Bubonic plague strikes London, killing approximately 100,000 people (20% of the city). The Lord Mayor orders watchmen to lock infected families inside houses marked with red crosses and 'Lord have mercy upon us'. Over 40,000 stray dogs and cats are slaughtered.

1665 • Robert Hooke Publishes Micrographia

Key Topic 2.1

Robert Hooke presents detailed microscopic drawings of fleas, cork cells, and fly eyes to the Royal Society. While microscopes prove the existence of an invisible world, lenses are too weak to reveal bacteria or germs, leaving medical treatments largely unchanged.

1676 • Thomas Sydenham Publishes Observationes Medicae

Key Topic 2.2

Known as the 'English Hippocrates', Thomas Sydenham urges doctors to base treatments on bedside clinical observation. In Observationes Medicae, he classifies illnesses into distinct species, distinguishes measles from scarlet fever, and popularises cinchona bark (quinine) for malaria.

How did new technologies and scientific societies challenge traditional medical authority?

AUDIO • QUIZ

1 c1440–1476 • Why did printing break the clerical monopoly on medical learning?

- Gutenberg printing press
- Movable metal type
- Rapid book mass-production

2 c1500 • How did humanism inspire physicians to challenge Galen?

- Humanism scholarly inquiry
- Original Greek translations
- Challenging Galenic dogma

3 1660 • Why was experimental observation central to the Royal Society?

- Royal Society founded (1660)
- Royal charter Charles II
- 'Nullius in Verba' motto

4 1665 • How did scientific journals speed up the exchange of medical knowledge?

- Philosophical Transactions (1665)
- Scientific paper peer-sharing
- Empirical laboratory debates

5 1665–1670s • Why did 17th-century microscopes fail to transform medical treatment?

- Compound light microscopes
- Robert Hooke Micrographia
- Leeuwenhoek 'animalcules'

Gutenberg gave the world movable type; the Royal Society gave the world peer review. Neither cured a cold, but both destroyed ignorance.

Question 4: Explain Why [12 marks • 15 mins]

Explain why new ideas in medicine were able to spread more rapidly in the period c1500–c1700. [12 marks]

Stimulus: **The invention of the printing press (c1440)** **The Royal Society (1660)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.1). In the drawing box, sketch Caxton's printing press beside the Royal Society's coat of arms (motto: Nullius in Verba). Annotate how printing removed the Church's control over book production. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [/ 4] • 5 MINS

Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]

Focus on persistent beliefs in miasma, supernatural forces, and humoral imbalance despite new anatomical knowledge.

Explain the mechanism: the Royal Society's journal (Philosophical Transactions) allowed European scientists to confirm and repeat experiments.

Why was Thomas Sydenham known as the 'English Hippocrates'?

AUDIO • QUIZ

1 1660s • Why did Sydenham believe bedside observation surpassed textbook learning?

- Thomas Sydenham bedside
- Close clinical observation
- Detailed symptom logging

2 1676 • How did classifying diseases challenge traditional individual humoral theory?

- Medical botany taxonomy
- Differential disease diagnosis
- Observationes Medicae (1676)

3 c1670s • Why was the success of Peruvian bark a mortal blow to humoral balance?

- Cinchona bark (quinine)
- Specific cures ague
- Challenging humour balance

4 c1670s • Why did Sydenham's cool treatment drastically cut smallpox deaths?

- Cool therapy treatment
- Fresh bedroom ventilation
- Discarding heavy blankets

5 1680s • Why did medical treatments remain mostly medieval despite Sydenham?

- Persistent miasma belief
- Pre-bacteriology limits
- Bleeding methods retained

Sydenham told medical students: "Go to the bedside, there alone can you learn disease." Textbooks are great, but patients actually have the symptoms.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why Thomas Sydenham was an important figure in the development of medicine in the seventeenth century. [12 marks]

Stimulus: **Observationes Medicae (1676)** | **Treatments for smallpox and malaria (cinchona bark)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.2). In the drawing box, sketch Sydenham taking clinical bedside notes beside a patient with cool fresh air flowing through an open window. Annotate why his classification of diseases was revolutionary. **[ ] Done**

Cool therapy for smallpox: opening windows and throwing off blankets. Radical in 1670, entirely sensible today.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the treatment of disease in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Focus on Sydenham's specific treatments (e.g. Peruvian bark for malaria) versus universal humoral purging.

Explain the impact: by proving cinchona bark cured only ague (malaria), Sydenham proved specific remedies could target specific diseases.

How did Andreas Vesalius revolutionize the study of human anatomy?

AUDIO • QUIZ

1 1537 • Why was a professor dissecting human bodies revolutionary in 1537?

- Andreas Vesalius Padua
- Direct human dissection
- Scalpel in hand

2 1540 • How did Vesalius prove Galen had made anatomical mistakes?

- Galenic errors corrected
- Single-bone human jaw
- Three-segment breastbone

3 1543 • Why was The Fabrica considered a masterpiece of Renaissance science?

- De Fabrica (1543)
- Master woodcut plates
- Johannes Oporinus printing

4 1545–1553 • How did English translations spread Vesalius's ideas to barber-surgeons?

- Thomas Geminus engravings
- English anatomy manual
- Barber-surgeon education

5 c1560 • Why did Vesalius's anatomical discoveries fail to save sick patients immediately?

- Conservative professor backlash
- Vesalius court resignation
- Empirical anatomy legacy

Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two. Awkward for Galenists!

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why Andreas Vesalius made significant breakthroughs in the understanding of human anatomy in the sixteenth century. [12 marks]

Stimulus: **De Humani Corporis Fabrica (1543)** | **Dissection of human corpses rather than animals** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.3). In the drawing box, sketch Vesalius standing over an anatomical dissection table holding a scalpel. Annotate two specific errors of Galen that he disproved. **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of anatomical training in the Renaissance (c1500–c1700) were similar to anatomical training in the nineteenth century. [4 marks]

Focus on direct physical dissection of human bodies as the foundation of surgical education.

Explain the limitation: while Vesalius proved Galen wrong on over 200 anatomical points, his discoveries did not immediately cure a single disease.

How did William Harvey prove the circulation of the blood, and why was it resisted?

AUDIO • QUIZ

1 c1615 • Why did Harvey study cold-blooded animals to understand the human heart?

- Girolamo Fabricius valves
- One-way blood flow
- Cold frog dissections

2 1620s • How did mathematical calculations prove Galen's theory impossible?

- Circulation mathematical proof
- Heart mechanical pump
- Galenic liver theory refuted

3 1628 • Why did Harvey publish his research abroad rather than in England?

- De Motu Cordis (1628)
- Forearm tight-ligature test
- Venous valve demonstration

4 1630s–1650s • Why was Harvey's discovery fiercely rejected by established doctors?

- Conservative medical opposition
- Losing private clients
- Slow university adoption

5 1661 • Why was the discovery of capillaries the final proof Harvey was right?

- Marcello Malpighi (1661)
- Microscopic capillary vessels
- Circulatory proof complete

Galen thought the liver made fresh blood like a factory; Harvey calculated you would need to produce 250 litres a day. Mathematics: the ultimate Galen-slayer.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why William Harvey's discovery of the circulation of the blood was a major breakthrough in medicine in the seventeenth century. [12 marks]

Stimulus: De Motu Cordis (1628) | Mechanical pump calculations and vein experiments | *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch Harvey's tied-arm ligature experiment showing blood pooling at the valves. Annotate his mathematical calculation proving Galen's liver-blood theory was impossible. **[ ] Done**

Harvey tied a bandage around an arm and watched the veins swell. Simple experiment, monumental consequence for human physiology.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the human body in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]

Contrast Harvey's dynamic mechanical circulation model with Galen's belief that the liver manufactured fresh blood consumed by organs.

Why did treatments during the Great Plague of 1665 show so much continuity with 1348?

AUDIO • QUIZ

1 Spring 1665 • Why did the rich flee London while the poor suffered catastrophic mortality?

- Great Plague (1665)
- Charles II court flight
- St Giles parish outbreak

2 Summer 1665 • Why did people still use 14th-century remedies in 1665?

- Wrath of God & miasma
- Plague water & pomanders
- Pigeon bubo remedies

3 1665 • How did local government action in 1665 show greater organization than 1348?

- Mayor quarantine orders
- Red cross marked doors
- Parish pest houses

4 1665 • Why was slaughtering cats and dogs a tragic medical mistake?

- Culling dogs & cats
- Street barrel fires
- Tobacco smoke warding

5 Winter 1665 • How devastating was the demographic impact of the 1665 epidemic?

- London dead carts
- 100,000 plague deaths
- Bills of Mortality records

1665: They killed 40,000 dogs and cats to stop the plague. The real culprits—the rat fleas—enjoyed a predator-free paradise.

Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

'Approaches to dealing with the Great Plague in 1665 were completely different from approaches to the Black Death in 1348.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]

Stimulus: **Mayor's plague orders and quarantine (red crosses)** | **Beliefs in miasma, astrology, and supernatural causes** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch a London front door marked with a red cross and guarded by a watchman. Annotate two continuities and two differences between 1348 and 1665. **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government responses to the Great Plague of London (1665) were different from government responses to the Black Death (1348). [4 marks]

Contrast systematic quarantine, red crosses, and pest houses in 1665 with rudimentary 1349 street cleaning.

KEY TOPIC 2 NOTES & EXAMINATION STRATEGY

STUDENT ASSESSMENT RECORD • KEY TOPIC 2 TRACKER

Paper 1: Medicine in Britain, c1250–present • Renaissance Britain (c1500–c1700)

PUPIL: _____ **TARGET GRADE:** ☐ **PREDICTED:** ☐ **ATTITUDE:** 1 • 2 • 3 • 4 • 5

#	ENQUIRY / SPECIFICATION FOCUS	DO NOW (10M)	Q3 SIM/DIFF (4M)	Q4/5 EXTENDED	LESSON TOTAL
1	**KT2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society** *"How did new technologies and scientific societies challenge traditional medical authority?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
2	**KT2.2: Thomas Sydenham & The Art of Bedside Observation** *"Why was Thomas Sydenham known as the 'English Hippocrates'?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
3	**KT2.3: Andreas Vesalius & The Anatomical Revolution** *"How did Andreas Vesalius revolutionize the study of human anatomy?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
4	**KT2.4: William Harvey & The Circulation of the Blood** *"How did William Harvey prove the circulation of the blood, and why was it resisted?"*	/ 10	/ 4	Q4 Explain: / 12	/ 26
5	**KT2.5: Continuity in Treatment & The Great Plague of London (1665)** *"Why did treatments during the Great Plague of 1665 show so much continuity with 1348?"*	/ 10	/ 4	Q5/6 Essay: / 20	/ 34
KEY TOPIC 2 CUMULATIVE ASSESSMENT TOTALS		**Do Now: / 50**	**Q3: / 20**	**Ext: / 68**	**/ 138**

TEACHER FORMATIVE ASSESSMENT • WWW / EBI FEEDBACK

WHAT WENT WELL (WWW):

_____

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EVEN BETTER IF (EBI):

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Teacher Signature: _____ **Date:** _____

DIGITAL QUIZZING HUB • 20-QUESTION PRACTICE QUIZZES

Scan with smartphone camera to open live interactive 20-question self-marking quizzes

KT 2.1	KT 2.2	KT 2.3	KT 2.4	KT 2.5
SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ	SCAN TO QUIZ
Score: / 20	**Score: / 20**	**Score: / 20**	**Score: / 20**	**Score: / 20**

Retain this student assessment record for parent consultations and GCSE revision monitoring.